

1. Executive Summary

This report outlines a comprehensive, end-to-end research plan for analyzing student perceptions of Generative AI (GenAI) in Data Science education. It integrates quantitative and qualitative NLP techniques—sentiment classification, summarization, thematic analysis, extractive QA, and an agentic QA assistant—while adhering to IEEE standards for research, software development, and ethics. The document critically examines each task’s feasibility, methodologies, required resources, and expected contributions to both theory and practice.

2. Introduction

2.1 Background. The rapid adoption of GenAI tools (e.g., ChatGPT, Bard) in Data Science coursework raises questions about student learning, confidence, ethics, and deep understanding. Prior studies have largely focused on AI tool efficacy or teacher perspectives; few center on detailed, student-generated reflections.

2.2 Research Problem. There is a lack of systematic, explainable frameworks to:

- Classify sentiment in student narratives.
- Summarize collective viewpoints per question.
- Uncover latent themes automatically.
- Provide interactive, extractive QA over open survey data.

2.3 Objectives.

1. Build sentiment classification for 100 open-ended responses.
 2. Generate abstractive summaries per question.
 3. Conduct automatic thematic analysis with LLM augmentation.
 4. Deploy a RAG-based extractive QA module.
 5. Develop an agentic QA assistant enabling dynamic user-driven exploration.
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3. Literature Review & Gap Analysis

Area	Key Works	Gap / Contribution
Sentiment Analysis	Liu et al. (2019), Hutto & Gilbert (VADER)	Domain-specific sentiment in GenAI education is underexplored.
Summarization	Zhang et al. (2020) PEGASUS; Lewis et al. (2021) BART	Few studies generate per-question student summaries at scale.

Area	Key Works	Gap / Contribution
Thematic Analysis	Braun & Clarke (2006) manual; Grootendorst (2022) BERTopic	Automated + LLM-assisted theme extraction for educational surveys is novel.
RAG QA Systems	Lewis et al. (2020) RAG; Karpukhin et al. (2020) DPR	Application of RAG to qualitative survey corpora remains scarce.
Agentic AI Assistants	Gao et al. (2023) ReAct	LLM agentic systems for pedagogical data are largely theoretical.

Critical Gap. Integrating these pipelines into a reproducible, IEEE-compliant framework for student feedback is an original contribution.

4. Data Description

4.1 Data Sources. CSV file with 100 student responses:

- 13 open-ended questions (text)
- 15 Likert-scale items (quantitative)

4.2 Data Characteristics.

- **Length Variability:** Responses range from 5 to 300 words.
- **Language:** English, with occasional domain jargon.
- **Balance:** Mixed sentiment across topics (preliminary manual scan).

4.3 Challenges.

- Small-sample risk for fine-tuning.
- Noise: typos, variable formatting.
- Domain-specific expressions ("I let AI handle code but struggled to debug").

5. Methodology

This section details each task with methods, standards, critical considerations, and deliverables.

5.1 Task 1: Sentiment Classification

5.1.1 Approach Options

- **Baseline:** VADER/TextBlob for polarity scores.
- **Zero-shot:** facebook/bart-large-mnli or GPT-4o prompt-based classification.
- **Supervised:** Fine-tune BERT on a manually labeled subset (~50 samples).

5.1.2 Process

1. Clean responses: lowercase, strip spaces.

2. Apply baseline and zero-shot; evaluate on manual labels.
3. If needed, annotate and fine-tune BERT (train/val/test splits).
4. Compute F1, precision, recall; human spot-check.

5.1.3 Critical Analysis

- **Feasibility:** High; zero-shot likely >70% accuracy.
- **Risk:** Low sample for fine-tuning.
- **Mitigation:** Use data augmentation or few-shot prompts.

5.1.4 Deliverables

- CSV with sentiment tags.
- Evaluation metrics and confusion matrix.

5.2 Task 2: Summarization per Question

5.2.1 Approach Options

- **Extractive:** TextRank / LexRank.
- **Abstractive:** `bart-large-cnn`, `flan-t5`, GPT-4o.

5.2.2 Process

1. Group responses by question.
2. If >4096 tokens, chunk into 500-word batches.
3. Summarize each chunk; merge and refine.
4. Evaluate via ROUGE scores and expert review (Likert readability).

5.2.3 Critical Analysis

- **Feasibility:** Very high; Colab handles model inference.
- **Risk:** Model hallucination or loss of nuance.
- **Mitigation:** Use human-in-the-loop refinement on key summaries.

5.2.4 Deliverables

- 13 paragraph summaries.
- ROUGE score table and readability ratings.

5.3 Task 3: Thematic Analysis (Automated)

5.3.1 Approach Options

- **LDA/NMF** on TF-IDF (baseline).
- **BERTopic** (recommended) for embedding-based clustering.
- **LLM labeling:** GPT-4o to name and describe themes.

5.3.2 Process

1. Preprocess: lemmatize, remove stop words.
2. Generate embeddings (`all-MiniLM-L6-v2`).
3. Cluster with UMAP + HDBSCAN → obtain topics.

4. Extract keywords per topic; obtain top 5 quotes.
5. Prompt GPT-4o: "Label these clusters and explain".

5.3.3 Critical Analysis

- **Feasibility:** High; BERTopic is robust on small corpora.
- **Risk:** Over-clustering or too granular themes.
- **Mitigation:** Tune HDBSCAN parameters; merge low-support topics.

5.3.4 Deliverables

- Theme list with labels, representative quotes.
- Coherence score and manual validation percentage.

5.4 Task 4: Extractive QA via RAG

5.4.1 Approach

- **Retriever:** FAISS index of chunk embeddings.
- **Reader:** `deepset/roberta-base-squad2` or GPT-4o span-extraction.

5.4.2 Process

1. Chunk responses (100–200 words).
2. Embed chunks; build FAISS index.
3. For an input query, retrieve top-5 chunks.
4. Pass to reader model; extract answer spans.
5. Rank and present top-3 spans with metadata.

5.4.3 Critical Analysis

- **Feasibility:** High; minimal compute.
- **Risk:** Context fragmentation; incorrect spans.
- **Mitigation:** Increase chunk overlap; rerank with LLM.

5.4.4 Deliverables

- API or notebook function: `answer_question(query)`.
- Example Q&A pairs.

5.5 Task 5: Agentic QA Assistant

5.5.1 Architecture

- **LangChain Agent:** Tools for retrieval, summarization, sentiment lookup.
- **Memory:** retain conversation context.
- **UI:** Streamlit web app.

5.5.2 Process

1. Define agent tools: `search_responses`, `summarize_question`, `get_sentiment`.
2. Initialize conversational agent using `ConversationalAgent` from LangChain.
3. Build Streamlit UI: input box, display functions.
4. Deploy on Streamlit Cloud or local Colab-driven UI.

5.5.3 Critical Analysis

- **Feasibility:** Moderate; requires chaining and error handling.
- **Risk:** Hallucinations or context drift.
- **Mitigation:** Restrict agent to defined tools; implement fallback responses.

5.5.4 Deliverables

- Live Streamlit app URL.
- README with usage examples.

6. Evaluation Plan (IEEE 730)

Task	Metric	Method
Sentiment	F1, Accuracy	Manual validation on 20% sample
Summaries	ROUGE-1/2/L, Readability	Expert Likert ratings
Themes	Coherence (c_v)	Compare BERTopic vs. LDA
QA	EM, F1 (SQuAD metrics)	Standard QA benchmarks, manual check
Agent	User satisfaction, Hallucination rate	Small user study, feedback survey

7. Ethical Considerations (IEEE 7000)

1. **Anonymization:** Remove any personally identifiable information.
2. **Bias Awareness:** Document limitations of LLM biases.
3. **Transparency:** Share code, parameters, random seeds.
4. **Consent:** Ensure students consented to use of their anonymized responses.

8. Project Timeline & Milestones

Phase	Milestone	Duration
Initialization	Data loading & cleaning complete	2 days
NLP Pipeline	Sentiment & summarization modules	4 days
Thematic Analysis	BERTopic + LLM labeling	3 days
RAG QA	Retriever + Reader integration	3 days
Agent & UI	LangChain agent + Streamlit app	5 days

Phase	Milestone	Duration
Evaluation & Reporting	Metrics computation & write-up	4 days
Total		21 days

9. Expected Contributions

- **Methodological:** A reusable, IEEE-compliant NLP framework for mixed-methods survey analysis.
 - **Empirical:** Detailed insights into student perceptions of GenAI in education.
 - **Technical:** Open-source Colab notebooks, Streamlit QA assistant.
 - **Theoretical:** Integrating thematic analysis with RAG and LLM agents.
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10. References

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End of Report